

# Unit-1

Random Variables and Probability Distributions

# Random variable

- A **random variable** is a function that associates a real number with each element in the sample space.
- **Example:**

Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values  $y$  of the random variable  $Y$ , where  $Y$  is the number of red balls, are

| Sample Space | $y$ |
|--------------|-----|
| $RR$         | 2   |
| $RB$         | 1   |
| $BR$         | 1   |
| $BB$         | 0   |



# Examples of random variable

- Example:

Statisticians use **sampling plans** to either accept or reject batches or lots of material. Suppose one of these sampling plans involves sampling independently 10 items from a lot of 100 items in which 12 are defective.

Let  $X$  be the random variable defined as the number of items found defective in the sample of 10. In this case, the random variable takes on the values  $0, 1, 2, \dots, 9, 10$ . 

- Example:

Let  $X$  be the random variable defined by the waiting time, in hours, between successive speeders spotted by a radar unit. The random variable  $X$  takes on all values  $x$  for which  $x \geq 0$ . 

# Types of random variable:

- If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers, it is called a **discrete sample space**
- If a sample space contains an infinite number of possibilities equal to the number of points on a line segment, it is called a **continuous sample space**.

# Probability mass function (p.m.f)

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The set of ordered pairs  $(x, f(x))$  is a **probability function**, **probability mass function**, or **probability distribution** of the discrete random variable  $X$  if, for each possible outcome  $x$ ,

1.  $f(x) \geq 0$ ,
2.  $\sum_x f(x) = 1$ ,
3.  $P(X = x) = f(x)$ .

The **cumulative distribution function**  $F(x)$  of a discrete random variable  $X$  with probability distribution  $f(x)$  is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad \text{for } -\infty < x < \infty.$$

# Discrete Probability distribution

**Q:** *A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.*

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

**Example 5.2.** A random variable  $X$  has the following probability distribution :

|          |   |     |      |      |      |       |        |            |
|----------|---|-----|------|------|------|-------|--------|------------|
| $x :$    | 0 | 1   | 2    | 3    | 4    | 5     | 6      | 7          |
| $p(x) :$ | 0 | $k$ | $2k$ | $2k$ | $3k$ | $k^2$ | $2k^2$ | $7k^2 + k$ |

(i) Find  $k$ , (ii) Evaluate  $P(X < 6)$ ,  $P(X \geq 6)$ , and  $P(0 < X < 5)$ , (iii) If  $P(X \leq c) > \frac{1}{2}$ , find the minimum value of  $c$ , and (iv) Determine the distribution function of  $X$ .

[Madurai Univ. B.Sc., Oct. 1988]

*Note :Also, you can discuss the  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

7. If  $p(x) = \frac{x}{15}; x = 1, 2, 3, 4, 5$   
 $= 0$ , elsewhere

Find (i)  $P\{X = 1 \text{ or } 2\}$ , and (ii)  $P\left\{\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right\}$

[Allahabad Univ. B.Sc., April 1992]

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $\text{Var}(X)$ ,  $E(aX+b)$ , and  $\text{Var}(aX+b)$  for the above question.*



# Practice problems

**Q:** Let  $W$  be a random variable giving the number of heads minus the number of tails in three tosses of a coin. List the elements of the sample space  $S$  for the three tosses of the coin and to each sample point assign a value  $w$  of  $W$ .

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

**Q:** Determine the value  $c$  so that each of the following functions can serve as a probability distribution of the discrete random variable  $X$ :

$$f(x) = c \binom{2}{x} \binom{3}{3-x}, \text{ for } x = 0, 1, 2.$$

**Q:** The probability distribution of  $X$ , the number of imperfections per 10 meters of a synthetic fabric in continuous rolls of uniform width, is given by

|        |      |      |      |      |      |
|--------|------|------|------|------|------|
| $x$    | 0    | 1    | 2    | 3    | 4    |
| $f(x)$ | 0.41 | 0.37 | 0.16 | 0.05 | 0.01 |

Construct the cumulative distribution function of  $X$ .

## Practice problems

Q: A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. If  $x$  is the number of defective sets purchased by the hotel, find the probability distribution of  $X$ .

Q: Three cards are drawn in succession from a deck without replacement. Find the probability distribution for the number of spades.

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

# Continuous Probability distribution

- The probability distribution of a continuous random variable cannot be presented in tabular form, it can be stated as a formula. Such a formula would necessarily be a function of the numerical values of the continuous random variable  $X$  and as such will be represented by the functional notation  $f(x)$ . In dealing with continuous variables,  $f(x)$  is usually called the probability density function, or simply the density function, of  $X$ .

The function  $f(x)$  is a **probability density function** (pdf) for the continuous random variable  $X$ , defined over the set of real numbers, if

1.  $f(x) \geq 0$ , for all  $x \in R$ .
2.  $\int_{-\infty}^{\infty} f(x) \, dx = 1$ .
3.  $P(a < X < b) = \int_a^b f(x) \, dx$ .

The **cumulative distribution function**  $F(x)$  of a continuous random variable  $X$  with density function  $f(x)$  is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt, \quad \text{for } -\infty < x < \infty.$$

**Example 5.4.** A continuous random variable  $X$  has a p.d.f.

$$f(x) = 3x^2, \quad 0 \leq x \leq 1. \text{ Find } a \text{ and } b \text{ such that}$$

(i)  $P\{X \leq a\} = P\{X > a\}$ , and

(ii)  $P\{X > b\} = 0.05$ .

**[Calicut Univ. B.Sc., Sept. 1988]**

(iii) Determine the median value (Note: The value of 'a' obtained in part (i) represents the median)"

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

Q: Suppose a certain type of small data processing firm is so specialized that some have difficulty making a profit in their first year of operation. The probability density function that characterizes the proportion  $Y$  that make a profit is given by

$$f(y) = \begin{cases} ky^4(1-y)^3, & 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) What is the value of  $k$  that renders the above a valid density function?
- (b) Find the probability that at most 50% of the firms make a profit in the first year.
- (c) Find the probability that at least 80% of the firms make a profit in the first year.

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

## Practice problems

Q) The diameter, say  $X$ , of an electric cable, is assumed to be a continuous random variable with p.d.f:  $f(x) = 6x(1-x)$ ,  $0 \leq x \leq 1$

- (i) Check that the above is a p.d.f
- (ii) Obtain an expression for c.d.f of  $X$ ,
- (iii) Compute  $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$ , and
- (iv) Determine the number  $k$  such that  $P(X < k) = P(X > k)$

Q) Let  $X$  be a continuous random variable with p.d.f:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0, & \text{else where} \end{cases}$$

Determine the constant  $a$ , and compute  $P(X \leq 1.5)$

Verify that the following is a distribution function

$$F(x) = \begin{cases} 0, & x < -a \\ \frac{1}{2} \left( \frac{x}{a} + 1 \right), & -a \leq x \leq a \\ 1, & x > a. \end{cases}$$

# Practice problems

Q: The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-8x}, & x \geq 0. \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (a) using the cumulative distribution function of X;
- (b) using the probability density function of X.

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*



# Practice problems

Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable  $X$  having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Verify that  $f(x)$  is a density function.
- (b) Find  $P(0 < X \leq 1)$ .
- (c) Find the Cumulative density function and use it to evaluate  $P(0 < X \leq 1)$ .

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

## Practice problems

Q: The proportion of people who respond to a certain mail-order solicitation is a continuous random variable  $X$  that has the density function

$$f(x) = \begin{cases} \frac{2(x+2)}{5}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Show that  $P(0 < X < 1) = 1$ .
- (b) Find the probability that more than 1/4 but fewer than 1/2 of the people contacted will respond to this type of solicitation.

Q: The shelf life, in days, for bottles of a certain prescribed medicine is a random variable having the density function

$$f(x) = \begin{cases} \frac{20,000}{(x+100)^3}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that a bottle of this medicine will have a shelf life of

- (a) at least 200 days;
- (b) anywhere from 80 to 120 days.

# Expected value of random variable

Let  $X$  be a random variable with probability distribution  $f(x)$ . The **mean**, or **expected value**, of  $X$  is

$$\mu = E(X) = \sum_x x f(x)$$

if  $X$  is discrete, and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

if  $X$  is continuous.

**Example 6.1.** Let  $X$  be a random variable with the following probability distribution :

|            |   |       |       |       |
|------------|---|-------|-------|-------|
| $x$        | : | $-3$  | $6$   | $9$   |
| $P_r(X=x)$ | : | $1/6$ | $1/2$ | $1/3$ |

Find  $E(X)$  and  $E(X^2)$  and using the laws of expectation, evaluate  $E(2X+1)^2$ .

**(Gauhati Univ. B.Sc., 1992)**

**Q:** A salesperson for a medical device company has two appointments on a given day. At the first appointment, he believes that he has a 70% chance to make the deal, from which he can earn \$1000 commission if successful. On the other hand, he thinks he only has a 40% chance to make the deal at the second appointment, from which, if successful, he can make \$1500. What is his expected commission based on his own probability belief? Assume that the appointment results are independent of each other.

# Practice problems

**Example 6.2.** (a) Find the expectation of the number on a die when thrown.

(b) Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.

**Example 6.5.** A coin is tossed until a head appears. What is the expectation of the number of tosses required ?  
[Delhi Univ. B.Sc., Oct. 1989]

Suppose that the number of cars  $X$  that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

| $x$        | 4              | 5              | 6             | 7             | 8             | 9             |
|------------|----------------|----------------|---------------|---------------|---------------|---------------|
| $P(X = x)$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

Let  $g(X) = 2X - 1$  represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

# Practice problems

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

**(ii) Find Variance of number of good components in sample.**

# Expected value of function of random variable

Let  $X$  be a random variable with probability distribution  $f(x)$ . The expected value of the random variable  $g(X)$  is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x)$$

if  $X$  is discrete, and

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) \, dx$$

if  $X$  is continuous.



# Expected value of function of two random variable

Let  $X$  and  $Y$  be random variables with joint probability distribution  $f(x, y)$ . The mean, or expected value, of the random variable  $g(X, Y)$  is

$$\mu_{g(X,Y)} = E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

if  $X$  and  $Y$  are discrete, and

$$\mu_{g(X,Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \, dx \, dy$$

if  $X$  and  $Y$  are continuous.

# Properties of expectation

- If  $a$  and  $b$  are constants, then

$$E(aX + b) = aE(X) + b.$$

- The expected value of the sum or difference of two or more functions of a random variable  $X$  is the sum or difference of the expected values of the functions. That is,

$$E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)].$$

- Let  $X$  and  $Y$  be two independent random variables. Then

$$E(XY) = E(X)E(Y).$$

- Let  $X$  and  $Y$  be two independent random variables. Then  $\sigma_{XY} = 0$ .

- If  $X$  and  $Y$  are random variables with joint probability distribution  $f(x, y)$  and  $a$ ,  $b$ , and  $c$  are constants, then

$$\sigma_{aX+bY+c}^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}.$$

## Practice problems

**4.11** The density function of coded measurements of the pitch diameter of threads of a fitting is

$$f(x) = \begin{cases} \frac{4}{\pi(1+x^2)}, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of  $X$ .

**4.20** A continuous random variable  $X$  has the density function

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of  $g(X) = e^{2X/3}$ .

**4.13** The density function of the continuous random variable  $X$ , the total number of hours, in units of 100 hours, that a family runs a vacuum cleaner over a period of one year, is given as

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the average number of hours per year that families run their vacuum cleaners.

# Variance of random variable

Let  $X$  be a random variable with probability distribution  $f(x)$  and mean  $\mu$ . The variance of  $X$  is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance,  $\sigma$ , is called the **standard deviation** of  $X$ .

The variance of a random variable  $X$  is

$$\sigma^2 = E(X^2) - \mu^2.$$

### Practice problems

4.34 Let  $X$  be a random variable with the following probability distribution:

|        |       |       |       |
|--------|-------|-------|-------|
| $x$    | $-2$  | $3$   | $5$   |
| $f(x)$ | $0.3$ | $0.2$ | $0.5$ |

Find the standard deviation of  $X$ .

4.50 For a laboratory assignment, if the equipment is working, the density function of the observed outcome  $X$  is

$$f(x) = \begin{cases} 2(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the variance and standard deviation of  $X$ .

# Joint probability distribution

If  $X$  and  $Y$  are two discrete random variables, the probability distribution for their simultaneous occurrence can be represented by a function with values  $f(x, y)$  for any pair of values  $(x, y)$  within the range of the random variables  $X$  and  $Y$ . It is customary to refer to this function as the **joint probability distribution** of  $X$  and  $Y$ .

Hence, in the discrete case,

$$f(x, y) = P(X = x, Y = y);$$

that is, the values  $f(x, y)$  give the probability that outcomes  $x$  and  $y$  occur at the same time. For example, if an 18-wheeler is to have its tires serviced and  $X$  represents the number of miles these tires have been driven and  $Y$  represents the number of tires that need to be replaced, then  $f(30000, 5)$  is the probability that the tires are used over 30,000 miles and the truck needs 5 new tires.

# Discrete Joint Probability Distribution

The function  $f(x, y)$  is a **joint probability distribution** or **probability mass function** of the discrete random variables  $X$  and  $Y$  if

1.  $f(x, y) \geq 0$  for all  $(x, y)$ ,
2.  $\sum_x \sum_y f(x, y) = 1$ ,
3.  $P(X = x, Y = y) = f(x, y)$ .

For any region  $A$  in the  $xy$  plane,  $P[(X, Y) \in A] = \sum \sum_A f(x, y)$ .

# Continuous joint probability distribution

The function  $f(x, y)$  is a **joint density function** of the continuous random variables  $X$  and  $Y$  if

1.  $f(x, y) \geq 0$ , for all  $(x, y)$ ,
2.  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$ ,
3.  $P[(X, Y) \in A] = \int \int_A f(x, y) \, dx \, dy$ , for any region  $A$  in the  $xy$  plane.



# Marginal probability distribution

The **marginal distributions** of  $X$  alone and of  $Y$  alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case, and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) \, dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) \, dx$$

for the continuous case.

# Conditional probability distribution

Let  $X$  and  $Y$  be two random variables, discrete or continuous. The **conditional distribution** of the random variable  $Y$  given that  $X = x$  is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0.$$

Similarly, the conditional distribution of  $X$  given that  $Y = y$  is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0.$$

# Statistically Independence

Let  $X$  and  $Y$  be two random variables, discrete or continuous, with joint probability distribution  $f(x, y)$  and marginal distributions  $g(x)$  and  $h(y)$ , respectively. The random variables  $X$  and  $Y$  are said to be **statistically independent** if and only if

$$f(x, y) = g(x)h(y)$$

for all  $(x, y)$  within their range.

# Covariance of random variable

Let  $X$  and  $Y$  be random variables with joint probability distribution  $f(x, y)$ . The covariance of  $X$  and  $Y$  is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if  $X$  and  $Y$  are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) \, dx \, dy$$

if  $X$  and  $Y$  are continuous.

The covariance of two random variables  $X$  and  $Y$  with means  $\mu_X$  and  $\mu_Y$ , respectively, is given by

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y.$$

**Example 6.32.** Let  $X$  and  $Y$  be two random variables each taking three values  $-1, 0$  and  $1$ , and having the joint probability distribution :

(i) Show that  $X$  and  $Y$  have different expectations.

| $Y \backslash X$ | $-1$          | $0$           | $1$           | Total         |
|------------------|---------------|---------------|---------------|---------------|
| $-1$             | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{3}{4}$ |
| $0$              | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ |
| $1$              | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{3}{4}$ |
| Total            | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{3}{4}$ | $\frac{9}{4}$ |

(ii) Prove that  $X$  and  $Y$  are uncorrelated.

(iii) Find  $\text{Var } X$  and  $\text{Var } Y$ .

(iv) Given that  $Y = 0$ , what is the conditional probability distribution of  $X$ ?

(v) Find  $V(Y | X = -1)$ .

Note: Part (ii) of **Example 6.32** is not relevant to Unit 1. You will be able to solve it after completing Unit 2

**3.43** Let  $X$  denote the reaction time, in seconds, to a certain stimulus and  $Y$  denote the temperature ( $^{\circ}\text{F}$ ) at which a certain reaction starts to take place. Suppose that two random variables  $X$  and  $Y$  have the joint density

$$f(x, y) = \begin{cases} 4xy, & 0 < x < 1, \ 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find

- (a)  $P(0 \leq X \leq \frac{1}{2} \text{ and } \frac{1}{4} \leq Y \leq \frac{1}{2})$ ;
- (b)  $P(X < Y)$ .

## Practice problems

**Example 5.20.** For the following bivariate probability distribution of  $X$  and  $Y$ , find

- (i)  $P(X \leq 1, Y = 2)$ , (ii)  $P(X \leq 1)$ , (iii)  $P(Y = 3)$ , (iv)  $P(Y \leq 3)$  and (v)  $P(X < 3, Y \leq 4)$

| $X \backslash Y$ | 1              | 2              | 3              | 4              | 5              | 6              |
|------------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 0                | 0              | 0              | $\frac{1}{32}$ | $\frac{2}{32}$ | $\frac{2}{32}$ | $\frac{3}{32}$ |
| 1                | $\frac{1}{16}$ | $\frac{1}{16}$ | $\frac{1}{8}$  | $\frac{1}{8}$  | $\frac{1}{8}$  | $\frac{1}{8}$  |
| 2                | $\frac{1}{32}$ | $\frac{1}{32}$ | $\frac{1}{64}$ | $\frac{1}{64}$ | 0              | $\frac{2}{64}$ |

The fraction  $X$  of male runners and the fraction  $Y$  of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

**3.49** Let  $X$  denote the number of times a certain numerical control machine will malfunction: 1, 2, or 3 times on any given day. Let  $Y$  denote the number of times a technician is called on an emergency call. Their joint probability distribution is given as

| $f(x, y)$ |   | $x$  |      |      |
|-----------|---|------|------|------|
|           |   | 1    | 2    | 3    |
| $y$       | 1 | 0.05 | 0.05 | 0.10 |
|           | 3 | 0.05 | 0.10 | 0.35 |
|           | 5 | 0.00 | 0.20 | 0.10 |

- Evaluate the marginal distribution of  $X$ .
- Evaluate the marginal distribution of  $Y$ .
- Find  $P(Y = 3 \mid X = 2)$ .

Find the covariance of  $X$  and  $Y$ .

### Practice problems

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find

- (a) the joint probability function  $f(x, y)$ ,
- (b)  $P[(X, Y) \in A]$ , where  $A$  is the region  $\{(x, y) | x + y \leq 1\}$ .

**3.38** If the joint probability distribution of  $X$  and  $Y$  is given by

$$f(x, y) = \frac{x + y}{30}, \quad \text{for } x = 0, 1, 2, 3; \ y = 0, 1, 2,$$

- find
- (a)  $P(X \leq 2, Y = 1)$ ;
  - (b)  $P(X > 2, Y \leq 1)$ ;
  - (c)  $P(X > Y)$ ;
  - (d)  $P(X + Y = 4)$ .



## Practice problems

Q) The diameter ,say  $X$  , of an electric cable, is assumed to be a continuous random variable with p.d.f:  $f(x) = 6x(1 - x), 0 \leq x \leq 1$

- (i) Check that the above is a p.d.f
- (ii) Obtain an expression for c.d.f of  $X$ ,
- (iii) Compute  $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$ , and
- (iv) Determine the number  $k$  such that  $P(X < k) = P(X > k)$

Q) Let  $X$  be a continuous random variable with p.d.f:

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 \leq x \leq 3 \\ 0, & \text{else where} \end{cases}$$

Determine the constant  $a$ , and compute  $P(X \leq 1.5)$

Verify that the following is a distribution function

$$F(x) = \begin{cases} 0, & x < -a \\ \frac{1}{2} \left( \frac{x}{a} + 1 \right), & -a \leq x \leq a \\ 1, & x > a. \end{cases}$$

*Note :Also, you can discuss the CDF,  $E(X)$ ,  $Var(X)$ ,  $E(aX+b)$ , and  $Var(aX+b)$  for the above question.*

## Practice problems

**Example 5.30.** If  $X$  and  $Y$  are two random variables having joint density function

$$f(x, y) = \frac{1}{8} (6 - x - y) ; 0 < x < 2, 2 < y < 4$$
$$= 0, \text{ otherwise}$$

Find (i)  $P(X < 1 \cap Y < 3)$ , (ii)  $P(X + Y < 3)$  and (iii)  $P(X < 1 | Y < 3)$   
(Madras Univ. B.Sc., Nov. 1986)

## Practice problems

**3.47** The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount  $Y$  from which a random amount  $X$  is sold during that day. Suppose that the tank is not resupplied during the day so that  $x \leq y$ , and assume that the joint density function of these variables is

$$f(x, y) = \begin{cases} 2, & 0 < x \leq y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Determine if  $X$  and  $Y$  are independent.
- (b) Find  $P(1/4 < X < 1/2 \mid Y = 3/4)$ .

**(iii)  $Var(X)$  and  $Var(Y)$ .**

**(iv) Co-variance between  $X$  and  $Y$ .**

## Practice problems

Q: Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. Two refills for a ballpoint pen are selected at random from a certain box, and the following is the joint probability distribution:

| $f(x, y)$ |   | $x$            |                 |                | $h(y)$          |
|-----------|---|----------------|-----------------|----------------|-----------------|
|           |   | 0              | 1               | 2              |                 |
| $y$       | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|           | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|           | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$    |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

Find the covariance of  $X$  and  $Y$ .

# Chebyshev's theorem

In probability theory, Chebyshev inequality guarantees, that for a wide class of probability distribution (Binomial, Poisson, normal,...), NO more than a certain fraction of values can be more than a certain distance from the mean.

Specifically, no more than  $1/(k^2)$  of the distribution values can be more than  $k$  standard deviation away from the mean (or equivalently, at least  $1 - 1/(k^2)$  of the distribution values are within  $k$  standard deviation of the mean)

# Chebyshev's theorem

:(Chebyshev's Theorem) The probability that any random variable  $X$  will assume a value within  $k$  standard deviations of the mean is at least  $1 - 1/k^2$ . That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

OR

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

Also for  $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

**Remark:-**

**By Chebyshev's Theorem**

- $\frac{3}{4}$  ( i.e. 75%) or more of the data of any distribution lies within two standard deviations of the mean i.e. in the interval  $(\mu \pm 2\sigma)$
- $\frac{8}{9}$  ( or 88.9%) or more of the data of any distribution lies within three standard deviations of the mean i.e. in the interval  $(\mu \pm 3\sigma)$
- $\frac{15}{16}$  ( or 93.75%) or more of the data of any distribution lies within four standard deviations of the mean i.e. in the interval  $(\mu \pm 4\sigma)$

.....

$(1 - \frac{1}{k^2})$  or more of the data of any distribution lies within  $k$  standard deviations of the mean i.e. in the interval  $(\mu \pm k\sigma)$

**4.77** A random variable  $X$  has a mean  $\mu = 10$  and a variance  $\sigma^2 = 4$ . Using Chebyshev's theorem, find

- (a)  $P(|X - 10| \geq 3)$ ;
- (b)  $P(|X - 10| < 3)$ ;
- (c)  $P(5 < X < 15)$ ;
- (d) the value of the constant  $c$  such that

$$P(|X - 10| \geq c) \leq 0.04.$$

**4.78** Compute  $P(\mu - 2\sigma < X < \mu + 2\sigma)$ , where  $X$  has the density function

$$f(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{elsewhere,} \end{cases}$$

and compare with the result given in Chebyshev's theorem.



## Practice problems

**Example 4.27:** A random variable  $X$  has a mean  $\mu = 8$ , a variance  $\sigma^2 = 9$ , and an unknown probability distribution. Find

- (a)  $P(-4 < X < 20)$ ,
- (b)  $P(|X - 8| \geq 6)$ .

Example : The average score in Mathematics exam is given as 84 with standard deviation 4. Find the range of marks between which at least 75% of the students scored.

**4.75** An electrical firm manufactures a 100-watt light bulb, which, according to specifications written on the package, has a mean life of 900 hours with a standard deviation of 50 hours. At most, what percentage of the bulbs fail to last even 700 hours? Assume that the distribution is symmetric about the mean.